

Divisor-Lifted Möbius Certificates on the Primitive Cofactor Plane: Two-Dimensional Abel Summation, Coprimality Tails, and the Physical Variation Gate

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

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Abstract

The cross-key erasure analysis at one fixed nonzero shift reduces the first genuinely signed native coefficient to a primitive cofactor-plane sum

$$\sum_{\substack{u \leq U, v \leq V \\ (u,v)=1}} \mu(u)\mu(v)K(u,v),$$

after complete native masks, prime-line fibers, and literal complex amplitudes have been retained in K . The sign is frozen along each prime-line parameter, so cancellation, if present, must occur across the (u, v) -plane.

We give an exact analytic certificate for this plane. First,

$$\mathbf{1}_{(u,v)=1}\mu(u)\mu(v) = \mu(uv).$$

Möbius inversion of the coprimality condition then yields the diagonal divisor lift

$$\sum_{d \geq 1} \mu(d) \sum_{\substack{a, b \geq 1 \\ (a,d)=(b,d)=1}} \mu(a)\mu(b)K(da, db).$$

For

$$M_d(x) = \sum_{\substack{n \leq x \\ (n,d)=1}} \mu(n),$$

two-dimensional Abel summation converts the d -term exactly into

$$\sum_{x,y} M_d(x)M_d(y)\Delta_{12}K^{[d]}(x,y).$$

Consequently, truncation at $d \leq D$ is controlled by restricted Mertens envelopes times zero-extended mixed variation, while the remaining diagonal tail is at most $\|K\|_\infty UV/D$.

The prime number theorem implies $M_d(x) = o(x)$ for every fixed d . It follows unconditionally that kernel families satisfying $\mathcal{V}_\square(K^{[d]}) = O_d(\|K\|_\infty)$ for every fixed d have $o(\|K\|_\infty UV)$ cancellation. In particular, the unweighted primitive rectangle cancels. We also prove projective bounded-variation and Banach-valued versions.

For the literal fixed-shift carrier, the excluded same-residual direction is one primitive-plane site and is already absent from the erasure kernel. The exact plane certificate bounds each anchored cross-key rectangle sum and hence the linear mixed trace $\text{tr}(ER)$. It is not an entrywise bound for the erasure operator. What remains unproved is stated without compression: mass-normalized variation for the actual prime-line kernel, a small diagonal tail at the physical scale, summability over the growing anchor family, an operator-level bridge, and the represented/missing shorted reassembly. Sign-selected hard masks show that bounded amplitude alone cannot replace these hypotheses. Thus the paper proves a regular-kernel cancellation theorem and isolates, but does not claim to solve, the fixed-shift parity interface.

Keywords: Möbius function; primitive cofactor plane; two-dimensional Abel summation; bounded variation; diagonal divisor lift; fixed shift; mixed-trace certificate.

Convention. The arithmetic specialization retains one fixed nonzero shift h_0 , complete native keys, the actual physical carrier, and raw atomic normalization; the shift is not specialized to 2.

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1 Introduction

The signed missing-native route separates two questions which are often blurred. The first is algebraic: after a residual label is erased, where does a nontrivial Möbius sign actually survive? TPC-68 shows that same-residual collisions freeze the target signs [6]. TPC-69 then expands the residual and erasure operators in noncommutative words and proves that the complete two-erasure sector in every even moment is nonnegative [7]. The first useful signed door is therefore the linear erasure sector or a higher, genuinely mixed sector.

TPC-70 opens the linear sector at the level of one two-key rectangle [8]. Its primitive coordinates have two decisive features. The common contents belong to divisor sets determined by $H = h_0(n - m)$, while the remaining erasure direction is a coprime pair (u, v) . More importantly, the surviving sign

$$\mu(a)\mu(b)\mu(u)\mu(v)$$

is constant along the affine time parameter of each prime ray. There is no time-randomness left to average on that ray. Any cancellation must join different primitive directions.

The present paper asks the next narrow question:

Which deterministic regularity of the exact primitive-plane kernel is sufficient to force cancellation, using only the prime number theorem?

The answer begins with a small identity that changes the geometry of the problem:

$$\mathbf{1}_{(u,v)=1}\mu(u)\mu(v) = \mu(uv).$$

The apparently rough coprimality boundary can be opened by a diagonal divisor lift. At each fixed divisor d , the resulting signs separate again into two copies of the Möbius function restricted to integers coprime to d . Two-dimensional Abel summation then transfers all cancellation into their partial sums and all geometric irregularity into the mixed variation of the dilated kernel.

1.1 Main results

Let K be supported on $[1, U] \times [1, V]$. Write

$$K^{[d]}(a, b) = K(da, db)$$

and extend it by zero outside its finite rectangle. If

$$\mathcal{V}_{\square}(K^{[d]}) = \sum_{a,b \geq 1} \left| K^{[d]}(a, b) - K^{[d]}(a+1, b) - K^{[d]}(a, b+1) + K^{[d]}(a+1, b+1) \right|,$$

and define

$$\mathfrak{M}_d(X) = \max_{1 \leq x \leq X} \left| \sum_{\substack{n \leq x \\ (n,d)=1}} \mu(n) \right|, \quad \mathcal{T}_D(K) \leq \|K\|_{\infty} UV/D.$$

Then the exact master bound proved below is

$$\left| \sum_{\substack{u \leq U, v \leq V \\ (u,v)=1}} \mu(u)\mu(v)K(u, v) \right| \leq \sum_{d \leq D} \mathfrak{M}_d(U/d) \mathfrak{M}_d(V/d) \mathcal{V}_{\square}(K^{[d]}) + \mathcal{T}_D(K), \quad (1)$$

The formula before absolute values is exact.

For fixed d , the prime number theorem gives $\mathfrak{M}_d(X) = o(X)$. Hence if

$$\mathcal{V}_{\square}(K^{[d]}) \ll_d \|K\|_{\infty}$$

for every fixed d , then the left side of (1) is

$$o(\|K\|_{\infty} UV)$$

as $\min(U, V) \rightarrow \infty$. This includes constant rectangles, finite projective sums of one-dimensional bounded-variation weights, and smooth rescaled kernels with uniformly bounded rectangular variation.

The result is deliberately normalized by ambient area. The physical prime-line kernel may be sparse, and its relevant denominator is an actual anchored rectangle mass rather than $\|K\|_{\infty} UV$. We therefore retain the right side of (1) as a literal majorant for each anchored sum and then sum those majorants in the exact trace formula. This produces three explicit arithmetic gates:

- (i) a mixed-variation bound after every low diagonal dilation;
- (ii) a diagonal-divisor tail small relative to actual physical mass; and
- (iii) summability over the growing row-and-anchor family at the physical trace scale.

None is silently identified with the smooth-kernel corollary. Moreover, a small mixed trace is not an operator bound. An entrywise, row-local, or growing-moment bridge remains a separate theorem.

1.2 What this does not prove

The exact physical kernel contains prime-line solvability, terminal masks, native-key coincidences, finite endpoints, and complex amplitudes. Those features can create many jumps in (u, v) . We do not prove that its mixed variation is bounded, or even smaller than its actual mass by a power. We also do not replace a row-pair estimate by a uniform operator estimate. Existing fixed-form and averaged Chowla technology does not automatically provide those missing interfaces; the coefficient and averaging mismatch was audited in TPC-33 [5].

Accordingly, the paper proves no fixed-2 estimate, no parity-breaking statement, and no prime-pair or twin-prime theorem. Its contribution is a positive theorem for regular primitive-plane kernels and an exact, falsifiable certificate for the irregular physical kernel.

1.3 Organization

Section 2 records the literal plane inherited from TPC-70. Section 3 proves the coprimality collapse and diagonal lift. Section 4 gives the exact Abel identity and restricted Mertens input. Section 5 proves the master truncation theorem. Section 6 develops unconditional regular-kernel consequences. Section 7 transfers the bound back to anchored and global mixed-trace certificates. Countermodels and the literature boundary follow before the final L0/L1/L2 ledger.

2 The primitive-plane interface

We first state the precise input inherited from TPC-70. This section is a dictionary, not a new estimate.

2.1 Anchor and erasure coordinates

Fix two distinct physical rows m, n , one nonzero shift h_0 , a same-residual anchor time j_R , and an unequal-residual erasure time j_E . We retain the literal hypotheses

$$(mn, h_0) = 1, \quad d_m \mid m, \quad d_n \mid n,$$

and all physical targets and residual labels used below are squarefree. Put

$$H = h_0(n - m), \quad \Delta = j_E - j_R.$$

At the anchor, write the two targets as

$$A_m = mj_R + h_0 = P_m R_m, \quad A_n = nj_R + h_0 = P_n R_n.$$

At the erasure time, write

$$B_m = mj_E + h_0 = Q_m S_m, \quad B_n = nj_E + h_0 = Q_n S_n.$$

The four exact determinant identities are

$$nA_m - mA_n = H, \quad nB_m - mB_n = H, \quad (2)$$

$$B_m - A_m = m\Delta, \quad B_n - A_n = n\Delta, \quad (3)$$

$$A_m B_n - B_m A_n = H\Delta. \quad (4)$$

Suppose that the anchor residuals agree after their row factors:

$$d_m R_m = d_n R_n.$$

Write

$$(d_m, d_n) = g, \quad d_m = ga, \quad d_n = gb, \quad (a, b) = 1.$$

On the squarefree physical carrier, g, a, b are pairwise coprime and squarefree. The equality forces

$$R_m = bt, \quad R_n = at$$

for a squarefree t coprime to gab , and the determinant equation forces

$$t \mid |H|.$$

For fixed t , the anchor prime factors lie on

$$nbP_m - maP_n = H/t.$$

At the erasure time, write

$$q = (S_m, S_n), \quad S_m = qu, \quad S_n = qv, \quad (u, v) = 1.$$

Then

$$q \mid |H|, \quad nuQ_m - mvQ_n = H/q.$$

The residuals are unequal exactly when

$$(u, v) \neq (b, a).$$

2.2 Frozen sign and the aggregate kernel

The physical target signs on the erasure edge reduce to

$$\mu(d_m S_m) \mu(d_n S_n) = \mu(a) \mu(b) \mu(u) \mu(v). \quad (5)$$

The common content q cancels, and the right side is constant along the affine prime ray that parametrizes the admissible erasure times for fixed (q, u, v) . Thus the time variable supplies weights and masks, not a new Möbius oscillation.

For a fixed same-residual anchor $\mathfrak{a}$, let

$$K_{\mathfrak{a}, q}(u, v)$$

denote the complete sum along that ray, including every literal terminal mask, endpoint, native-key constraint, prime-factor condition, normalization, and complex amplitude, and summing only actual

unequal-residual erasure keys completing $\mathfrak{a}$. It is defined to be zero when the corresponding ray is absent. In particular,

$$K_{\mathfrak{a},q}(b, a) = 0.$$

Aggregate the divisor-many contents before taking absolute values:

$$K_{\mathfrak{a}}(u, v) = \sum_{q \parallel H} K_{\mathfrak{a},q}(u, v). \quad (6)$$

Thus the literal aggregate is already zero at the excluded direction; set

$$K_{\mathfrak{a}}^{\circ}(u, v) = K_{\mathfrak{a}}(u, v). \quad (7)$$

If one instead introduces an enlarged bookkeeping kernel $\widehat{K}_{\mathfrak{a}}$ that includes the candidate site (b, a) , the corresponding literal kernel is

$$K_{\mathfrak{a}}^{\circ} = \widehat{K}_{\mathfrak{a}} - \widehat{K}_{\mathfrak{a}}(b, a)e_{b,a}.$$

That is an optional analytic convention, not an extra physical atom. The literal signed sum attached to $\mathfrak{a}$ is then

$$\boxed{\mathfrak{X}_{\mathfrak{a}} = \mu(a)\mu(b) \sum_{\substack{u,v \geq 1 \\ (u,v)=1}} \mu(u)\mu(v)K_{\mathfrak{a}}^{\circ}(u, v).} \quad (8)$$

If $\mathcal{R}_{m,n}$ is the set of actual same-residual anchors for $m < n$, TPC-70 gives

$$\mathrm{tr}(ER) = 2 \operatorname{Re} \sum_{m < n} \sum_{\mathfrak{a} \in \mathcal{R}_{m,n}} \mathfrak{X}_{\mathfrak{a}}. \quad (9)$$

The normalization $W_m W_n$ is already contained in every literal four-corner amplitude inside $K_{\mathfrak{a},q}$.

Remark 2.1 (Aggregation order). The sum over $q \mid |H|$ in (6) precedes all absolute values. Bounding each q -slice separately is valid, but may pay an unnecessary divisor factor and discard literal phase cancellation.

Remark 2.2 (Anchor and operator scope). The primitive coordinates a, b, u, v depend on the row pair and the kernel depends on the complete same-residual anchor. A theorem for one $\mathfrak{a}$, or even cancellation after summing all anchors in (9), is not an entrywise theorem for E and is not an operator theorem. This distinction is retained in [section 7](#).

3 Coprimality collapse and the diagonal divisor lift

Throughout this section, $K : \mathbb{N}^2 \rightarrow \mathbb{C}$ has finite support. All sums are therefore finite, and no limiting interchange is hidden.

Lemma 3.1 (Primitive sign collapse). *For all positive integers u, v ,*

$$\boxed{\mathbf{1}_{(u,v)=1} \mu(u)\mu(v) = \mu(uv).} \quad (10)$$

Proof. If either u or v is not squarefree, both sides vanish. If both are squarefree and coprime, multiplicativity gives $\mu(uv) = \mu(u)\mu(v)$. If both are squarefree but share a prime, then uv is not squarefree, so both sides again vanish. $\square$

The identity is elementary, but its role is structural. The coprimality mask and the two signs are not three separate objects; together they are one product Möbius coefficient.

Lemma 3.2 (Coprimality inversion). *For all $u, v \geq 1$,*

$$\mu(uv) = \mu(u)\mu(v) \sum_{d \mid (u,v)} \mu(d). \quad (11)$$

Proof. If $\mu(u)\mu(v) = 0$, both sides vanish. Otherwise u and v are squarefree, and

$$\sum_{d|(u,v)} \mu(d) = \mathbf{1}_{(u,v)=1}.$$

Apply [lemma 3.1](#). □

Theorem 3.3 (Exact diagonal divisor lift). *Define*

$$\mathcal{S}(K) = \sum_{\substack{u,v \geq 1 \\ (u,v)=1}} \mu(u)\mu(v)K(u,v).$$

Then

$$\mathcal{S}(K) = \sum_{u,v \geq 1} \mu(uv)K(u,v) \tag{12}$$

$$= \sum_{d \geq 1} \mu(d) \sum_{\substack{x,y \geq 1 \\ (x,d)=(y,d)=1}} \mu(x)\mu(y)K(dx, dy). \tag{13}$$

Proof. The first equality is [lemma 3.1](#). Insert (11) and interchange finite sums:

$$\mathcal{S}(K) = \sum_{d \geq 1} \mu(d) \sum_{x,y \geq 1} \mu(dx)\mu(dy)K(dx, dy).$$

Only squarefree d contribute. For such d ,

$$\mu(dx) = \mu(d)\mu(x)\mathbf{1}_{(x,d)=1},$$

with the same identity for y . The two factors $\mu(d)$ square to one, leaving (13). □

Remark 3.4 (The diagonal variable is new). The divisor d in (13) is introduced by coprimality inversion. It is not the row cofactor d_m , the anchor content t , or the erasure content q of [section 2](#). Conflating these variables would change both the support and the sign.

Remark 3.5 (Why the lift helps). For a fixed d , the two variable signs in (13) separate. Their only modification is the fixed local condition $(x,d) = (y,d) = 1$. The price is a sum over diagonal dilations $K(dx, dy)$. Low d will be handled by the prime number theorem; large d will be an explicit tail.

3.1 Removing one primitive direction

The physical erasure plane excludes the anchor direction (b, a) , where $(a, b) = 1$. For any kernel K , define

$$K^\circ = K - K(b, a)e_{b,a}, \quad e_{b,a}(u, v) = \mathbf{1}_{(u,v)=(b,a)}.$$

Then

$$\mathcal{S}(K^\circ) = \mathcal{S}(K) - \mu(a)\mu(b)K(b, a). \tag{14}$$

Thus the unequal-residual condition is exactly one primitive-plane site correction. Its value may aggregate several underlying labels and is not called one physical atom. For the literal TPC-70 erasure kernel the site is already zero; the formula is useful only if one first works with an enlarged bookkeeping kernel.

Lemma 3.6 (Diagonal visibility of the removed point). *Let $K^{[d]}(x, y) = K(dx, dy)$. The removed point (b, a) appears in $(K^\circ)^{[d]}$ only for $d = 1$. For every $d > 1$,*

$$(K^\circ)^{[d]} = K^{[d]},$$

and hence

$$\mathcal{V}_\square((K^\circ)^{[d]}) = \mathcal{V}_\square(K^{[d]})$$

For $d = 1$,

$$\mathcal{V}_\square(K^\circ) \leq \mathcal{V}_\square(K) + 4|K(b, a)|.$$

Proof. The point survives the d -dilation only if $d \mid a$ and $d \mid b$. Since $(a, b) = 1$, this forces $d = 1$. A point mass has four nonzero mixed differences, counted with multiplicity at the axes when necessary, and hence mixed variation at most four times its amplitude. □

4 Restricted Mertens sums and two-dimensional Abel summation

4.1 Zero-extended differences

For a finitely supported kernel $F : \mathbb{N}^2 \rightarrow \mathbb{C}$, set

$$\begin{aligned}\Delta_1 F(x, y) &= F(x, y) - F(x + 1, y), \\ \Delta_2 F(x, y) &= F(x, y) - F(x, y + 1), \\ \Delta_{12} F(x, y) &= F(x, y) - F(x + 1, y) - F(x, y + 1) + F(x + 1, y + 1).\end{aligned}$$

All values outside the declared support are zero. Define the rectangular mixed variation

$$\mathcal{V}_{\square}(F) = \sum_{x, y \geq 1} |\Delta_{12} F(x, y)|. \quad (15)$$

Because of zero extension, $\mathcal{V}_{\square}$ includes all upper and right boundary jumps and the terminal corner. No separate edge term is omitted.

Lemma 4.1 (Two-dimensional Abel identity). *Let (α_x) and (β_y) be finitely supported sequences, and put*

$$A(x) = \sum_{r \leq x} \alpha_r, \quad B(y) = \sum_{s \leq y} \beta_s.$$

For every finitely supported F ,

$$\boxed{\sum_{x, y \geq 1} \alpha_x \beta_y F(x, y) = \sum_{x, y \geq 1} A(x) B(y) \Delta_{12} F(x, y).} \quad (16)$$

Proof. One-dimensional summation by parts with zero extension gives

$$\sum_x \alpha_x G(x) = \sum_x A(x) (G(x) - G(x + 1)).$$

Apply this first in x for each y , and then in y . The resulting second difference is exactly $\Delta_{12} F$. $\square$

Remark 4.2 (Equivalent edge–corner form). If a kernel is specified only on a closed rectangle and not zero-extended, expanding (16) produces an interior mixed-difference sum, two terminal edge sums, and one corner term. Definition (15) packages all four pieces without losing them.

4.2 Restricted Mertens envelopes

For $d \geq 1$, define

$$M_d(x) = \sum_{\substack{n \leq x \\ (n, d) = 1}} \mu(n), \quad \mathfrak{M}_d(X) = \max_{1 \leq x \leq X} |M_d(x)|. \quad (17)$$

We set $\mathfrak{M}_d(X) = 0$ when $X < 1$. These quantities depend only on the squarefree kernel of d :

$$M_d = M_{\text{rad}(d)}.$$

We use only the classical prime-number-theorem consequence

$$M_1(x) = o(x);$$

see, for example, Iwaniec and Kowalski [2, Chapter 5].

Lemma 4.3 (Fixed local factors preserve Mertens cancellation). *For every fixed positive integer d ,*

$$M_d(x) = o_d(x), \quad \mathfrak{M}_d(X) = o_d(X). \quad (18)$$

Proof. It suffices to adjoin one prime at a time. If $p \nmid d$, partition the sum defining $M_d(x)$ according to divisibility by p . For the p -divisible terms, write $n = pr$. A nonzero term requires $(r, dp) = 1$, and then $\mu(pr) = -\mu(r)$. Therefore

$$M_{dp}(x) = M_d(x) + M_{dp}(x/p).$$

Iteration gives

$$M_{dp}(x) = \sum_{k \geq 0} M_d(x/p^k), \quad (19)$$

where the sum is finite.

If $M_d(t) = o(t)$, split (19) at a fixed large threshold. Above the threshold, the geometric sum is $o(x)$; below it, the trivial estimate $|M_d(t)| \leq t$ gives a bounded geometric tail, also $o(x)$. Starting with $d = 1$ proves the first claim by induction over the prime divisors of d .

For the maximal claim, given $\varepsilon > 0$, take x_0 such that $|M_d(x)| \leq \varepsilon x$ for $x \geq x_0$. Then

$$\mathfrak{M}_d(X) \leq \max_{x < x_0} |M_d(x)| + \varepsilon X$$

Divide by X , take the limit superior, and then let $\varepsilon \rightarrow 0$. This proves $\mathfrak{M}_d(X) = o(X)$. $\square$

Remark 4.4 (No growing- d uniformity). [Lemma 4.3](#) holds for each fixed d . It does not state a uniform estimate when d grows with the physical scale. The master certificate below avoids silently assuming such uniformity by truncating at fixed D and retaining the remainder as an explicit tail.

Proposition 4.5 (Abel form of one diagonal slice). *For*

$$K^{[d]}(x, y) = K(dx, dy),$$

one has

$$\begin{aligned} \sum_{\substack{x, y \geq 1 \\ (x, d) = (y, d) = 1}} \mu(x)\mu(y)K(dx, dy) \\ = \sum_{x, y \geq 1} M_d(x)M_d(y)\Delta_{12}K^{[d]}(x, y). \end{aligned} \quad (20)$$

Consequently,

$$\left| \sum_{\substack{x, y \geq 1 \\ (x, d) = (y, d) = 1}} \mu(x)\mu(y)K(dx, dy) \right| \leq \mathfrak{M}_d(U/d)\mathfrak{M}_d(V/d)\mathcal{V}_{\square}(K^{[d]}) \quad (21)$$

when K is supported on $[1, U] \times [1, V]$.

Proof. Apply [lemma 4.1](#) to

$$\alpha_x = \mu(x)\mathbf{1}_{(x, d)=1}, \quad \beta_y = \mu(y)\mathbf{1}_{(y, d)=1}.$$

Although these sequences are infinite, the kernel is supported on a finite rectangle; truncate both sequences beyond that rectangle before applying the lemma. Their partial sums are $M_d(x)$ and $M_d(y)$. $\square$

5 The master divisor–variation certificate

Let K be supported on $[1, U] \times [1, V]$. For $D \geq 1$, define the exact diagonal tail

$$\mathcal{T}_D(K) = \sum_{d > D} |\mu(d)| \sum_{x, y \geq 1} |\mu(dx)\mu(dy)K(dx, dy)|. \quad (22)$$

This definition retains the actual support and coefficients. It can be substantially smaller than the ambient estimate used below.

Theorem 5.1 (Exact truncated expansion and master bound). *For every integer $D \geq 1$,*

$$\mathcal{S}(K) = \sum_{d \leq D} \mu(d) \sum_{x, y \geq 1} M_d(x) M_d(y) \Delta_{12} K^{[d]}(x, y) + \mathcal{R}_D(K), \quad (23)$$

$$|\mathcal{R}_D(K)| \leq \mathcal{T}_D(K). \quad (24)$$

In particular,

$$\boxed{|\mathcal{S}(K)| \leq \sum_{d \leq D} \mathfrak{M}_d(U/d) \mathfrak{M}_d(V/d) \mathcal{V}_{\square}(K^{[d]}) + \mathcal{T}_D(K).} \quad (25)$$

Proof. Split the exact diagonal lift (13) at $d \leq D$. Apply [proposition 4.5](#) to every low diagonal. The remaining exact sum is $\mathcal{R}_D(K)$; the triangle inequality gives (24). A second triangle inequality and (21) give (25). $\square$

Proposition 5.2 (Ambient diagonal-tail estimate). *For every $D \geq 1$,*

$$\boxed{\mathcal{T}_D(K) \leq \|K\|_{\infty} \sum_{d > D} \left\lfloor \frac{U}{d} \right\rfloor \left\lfloor \frac{V}{d} \right\rfloor \leq \frac{\|K\|_{\infty} UV}{D}.} \quad (26)$$

Proof. Drop the Möbius factors in (22). For fixed d , at most $\lfloor U/d \rfloor \lfloor V/d \rfloor$ pairs occur. Finally,

$$\sum_{d > D} \frac{1}{d^2} \leq \frac{1}{D}.$$

$\square$

Remark 5.3 (Order of limits). The safe unconditional procedure is: first fix D , use [lemma 4.3](#) on the finitely many low diagonals, take the physical scale to infinity, and only then let $D \rightarrow \infty$. Choosing a growing D at the start would require a new uniform estimate for M_d .

5.1 A mass-normalized form

For applications, choose a positive reference scale $\mathcal{A}(K)$. It may be the actual ℓ^1 atomic mass, a row-pair majorant, or a normalized Gram denominator. Define

$$\mathcal{B}_D(K) = \sum_{d \leq D} \mathfrak{M}_d(U/d) \mathfrak{M}_d(V/d) \mathcal{V}_{\square}(K^{[d]}) + \mathcal{T}_D(K). \quad (27)$$

Then

$$\frac{|\mathcal{S}(K)|}{\mathcal{A}(K)} \leq \frac{\mathcal{B}_D(K)}{\mathcal{A}(K)} \quad (28)$$

whenever $\mathcal{A}(K) > 0$. This tautological-looking normalization is essential. A bound $o(\|K\|_{\infty} UV)$ can be useless if the physical carrier has only $o(UV)$ actual mass.

Definition 5.4 (Variation–tail gate). A family K_X closes the variation–tail gate at reference scale $\mathcal{A}_X$ if there is a choice of $D = D(X)$, justified with whatever uniform restricted-Mertens input it uses, such that

$$\mathcal{B}_{D(X)}(K_X) = o(\mathcal{A}_X).$$

With fixed- d input only, a sufficient safe formulation is a two-limit statement: low-diagonal contributions are $o_D(\mathcal{A}_X)$ for each fixed D , and

$$\lim_{D \rightarrow \infty} \limsup_{X \rightarrow \infty} \frac{\mathcal{T}_D(K_X)}{\mathcal{A}_X} = 0.$$

Remark 5.5. The definition is a sufficient certificate, not a necessary condition for cancellation. Large variation can make this route fail even when the original signed sum is small for another reason.

5.2 Banach-valued version

The exact formulas also hold for kernels taking values in a complex Banach space X . Replace absolute values of mixed differences by X -norms and define

$$\mathcal{V}_{\square, \mathsf{X}}(K) = \sum_{x, y} \|\Delta_{12}K(x, y)\|_{\mathsf{X}}.$$

Corollary 5.6 (Banach-valued certificate). *For finitely supported $K : \mathbb{N}^2 \rightarrow \mathsf{X}$,*

$$\|\mathcal{S}(K)\|_{\mathsf{X}} \leq \sum_{d \leq D} \mathfrak{M}_d(U/d) \mathfrak{M}_d(V/d) \mathcal{V}_{\square, \mathsf{X}}(K^{[d]}) + \mathcal{T}_{D, \mathsf{X}}(K),$$

where the tail is defined by replacing scalar absolute values in (22) with X -norms.

Proof. All sums are finite. The Abel identities use only addition and scalar multiplication; the final estimate is the triangle inequality in X . $\square$

Remark 5.7. Taking X to be a matrix space with operator or star norm can avoid an entrywise loss when a common primitive coordinate system exists. In the literal carrier, however, the primitive coordinates depend on the row pair. That alignment issue must be verified rather than assumed.

5.3 An anisotropic one-direction certificate

The full mixed-variation norm is not the only option. If one cofactor direction is regular while the other is rough, perform Abel summation only in the regular direction and take absolute values in the other.

Definition 5.8. For a finitely supported kernel F , set

$$\mathcal{V}_{1, \ell^1}(F) = \sum_{x, y \geq 1} |\Delta_1 F(x, y)|, \quad \mathcal{V}_{2, \ell^1}(F) = \sum_{x, y \geq 1} |\Delta_2 F(x, y)|.$$

Zero extension again includes terminal boundary jumps.

Proposition 5.9 (One-direction diagonal certificate). *For every $D \geq 1$,*

$$|\mathcal{S}(K)| \leq \sum_{d \leq D} \mathfrak{M}_d(U/d) \mathcal{V}_{1, \ell^1}(K^{[d]}) + \mathcal{T}_D(K), \quad (29)$$

$$|\mathcal{S}(K)| \leq \sum_{d \leq D} \mathfrak{M}_d(V/d) \mathcal{V}_{2, \ell^1}(K^{[d]}) + \mathcal{T}_D(K). \quad (30)$$

Proof. In the low- d term of (13), fix y and apply one-dimensional summation by parts in x . Its absolute value is at most

$$\mathfrak{M}_d(U/d) \sum_x |\Delta_1 K^{[d]}(x, y)|.$$

Sum over y , using $|\mu(y) \mathbf{1}_{(y, d)=1}| \leq 1$, then sum over $d \leq D$ and retain the same exact tail. The other inequality is symmetric. $\square$

Corollary 5.10 (One regular direction is enough at ambient scale). *Suppose $U \rightarrow \infty$, $B = \|K\|_{\infty} > 0$, and, for every fixed d ,*

$$\mathcal{V}_{1, \ell^1}(K^{[d]}) \ll_d BV.$$

Then

$$\mathcal{S}(K) = o(BUV)$$

provided the ambient diagonal tail is treated by taking the scale limit first and then $D \rightarrow \infty$. A symmetric statement holds when only $V \rightarrow \infty$.

Proof. Use (29), $\mathfrak{M}_d(U/d) = o_d(U/d)$, and (26). $\square$

Remark 5.11. The anisotropic certificate gains only one Mertens factor, but it does not charge variation in the rough direction. It is therefore a genuine alternative gate, not merely a weakening of notation.

6 Unconditional cancellation for regular plane kernels

The master certificate has a positive unconditional consequence. It is useful both as a theorem and as a calibration of what the physical kernel would need to resemble.

Theorem 6.1 (Bounded fixed-d variation implies cancellation). *Let $K_{U,V}$ be supported on $[1, U] \times [1, V]$, and put*

$$B_{U,V} = \|K_{U,V}\|_\infty.$$

Assume $B_{U,V} > 0$ and, for every fixed $d \geq 1$,

$$\mathcal{V}_\square(K_{U,V}^{[d]}) \leq C_d B_{U,V} \quad (31)$$

with C_d independent of U, V . Then, as $\min(U, V) \rightarrow \infty$,

$$\boxed{\sum_{\substack{u \leq U, v \leq V \\ (u,v)=1}} \mu(u)\mu(v)K_{U,V}(u, v) = o(B_{U,V}UV)}. \quad (32)$$

Proof. Fix D . By lemma 4.3, for every $d \leq D$,

$$\mathfrak{M}_d(U/d)\mathfrak{M}_d(V/d) = o_d(UV/d^2).$$

The finitely many low-diagonal terms in (25), divided by $B_{U,V}UV$, therefore tend to zero by (31). The ambient tail (26) contributes at most $1/D$. Take the scale limit first and then $D \rightarrow \infty$. $\square$

Corollary 6.2 (The primitive rectangle). *As $\min(U, V) \rightarrow \infty$,*

$$\sum_{\substack{u \leq U, v \leq V \\ (u,v)=1}} \mu(u)\mu(v) = o(UV). \quad (33)$$

Proof. Take $K = \mathbf{1}_{[1,U] \times [1,V]}$. For every d , its zero-extended dilation is another rectangle and

$$\mathcal{V}_\square(K^{[d]}) = 1.$$

Apply theorem 6.1. $\square$

Remark 6.3. Corollary 6.2 is not the two-point Chowla conjecture. The two variables are independently averaged over a full rectangle, and the coprimality condition turns the sign into $\mu(uv)$. No prescribed affine relation such as $v = u + 1$ is imposed.

6.1 Projective bounded variation

For a finitely supported one-dimensional sequence f , define its zero-extended variation

$$\mathcal{V}_0(f) = \sum_{x \geq 1} |f(x) - f(x+1)|.$$

Lemma 6.4 (Projective variation bound). *If*

$$K(x, y) = \sum_{r=1}^R f_r(x)g_r(y),$$

then

$$\mathcal{V}_\square(K) \leq \sum_{r=1}^R \mathcal{V}_0(f_r)\mathcal{V}_0(g_r). \quad (34)$$

Proof. For a rank-one kernel,

$$\Delta_{12}(fg)(x, y) = (f(x) - f(x+1))(g(y) - g(y+1)).$$

Sum absolute values and then use the triangle inequality over r . $\square$

Corollary 6.5 (Finite projective-BV families). *Suppose that, for every fixed d , the dilated kernels admit representations*

$$K_{U,V}^{[d]}(x, y) = \sum_r f_{r,d}(x) g_{r,d}(y)$$

such that

$$\sum_r \mathcal{V}_0(f_{r,d}) \mathcal{V}_0(g_{r,d}) \ll_d B_{U,V}.$$

Then (32) holds.

Proof. Combine lemma 6.4 with theorem 6.1. □

6.2 Dyadic localization

The same argument applies to translated rectangles. If K is supported on

$$U_0 < u \leq U_1, \quad V_0 < v \leq V_1,$$

zero extension automatically inserts the four boundary jumps. One may also use interval partial sums

$$M_d(x) - M_d(U_0/d)$$

inside the Abel formula. On dyadic boxes with $U, V \rightarrow \infty$,

$$U < u \leq 2U, \quad V < v \leq 2V,$$

fixed- d Mertens cancellation is still $o(U)$ and $o(V)$. Therefore theorem 6.1 is stable under a bounded number of dyadic localizations. Summing $O(\log^2 X)$ boxes is harmless only when the target ledger permits that subpower bookkeeping and when the variation bounds are uniform over the boxes. Boxes with $U = O(1)$ or $V = O(1)$ belong instead to the boundary or anisotropic sectors of section 7.

6.3 A squarefree-pair calibration

The set on which $\mu(uv) \neq 0$ has positive planar density

$$\mathfrak{S}_{\text{sf}} = \prod_p \left(1 - \frac{3}{p^2} + \frac{2}{p^3} \right) > 0. \quad (35)$$

Indeed, the local condition is that the sum of the two p -adic valuations is at most one. Its local density is

$$(1 - p^{-1})^2 + 2(p^{-1} - p^{-2})(1 - p^{-1}) = 1 - \frac{3}{p^2} + \frac{2}{p^3}.$$

For a fixed z , the Chinese remainder theorem therefore gives the product of these factors over $p \leq z$, up to $O_z(U + V + 1)$ boundary pairs. A violation caused by a prime $p > z$ requires

$$p^2 \mid u, \quad \text{or} \quad p^2 \mid v, \quad \text{or} \quad p \mid (u, v).$$

The union of these events contains at most

$$3UV \sum_{p > z} \frac{1}{p^2}$$

pairs. Let first $\min(U, V) \rightarrow \infty$ and then $z \rightarrow \infty$. This proves

$$\sum_{u \leq U, v \leq V} \mu(uv)^2 = \mathfrak{S}_{\text{sf}} UV + o(UV) \quad (36)$$

as $\min(U, V) \rightarrow \infty$. Together with corollary 6.2, this shows that the signs $+1$ and -1 of $\mu(uv)$ each occupy asymptotically $\mathfrak{S}_{\text{sf}} UV/2$ points. This fact will calibrate the hard-mask countermodel in section 8.

7 Return to the fixed-shift mixed trace

We now apply the deterministic theorem to the exact anchored kernel in (8). This section states certificates, not estimates for the physical kernel. The object being bounded is kept explicit: it is an anchored four-corner scalar contributing to $\text{tr}(ER)$, not a two-corner matrix entry of E .

7.1 One same-residual anchor

Assume that K_a° is supported on $[1, U_a] \times [1, V_a]$. Define

$$\begin{aligned} \mathcal{B}_a(D) &= \sum_{d \leq D} \mathfrak{M}_d(U_a/d) \mathfrak{M}_d(V_a/d) \mathcal{V}_\square((K_a^\circ)^{[d]}) \\ &\quad + \mathcal{T}_D(K_a^\circ). \end{aligned} \tag{37}$$

Proposition 7.1 (Literal anchored certificate). *For every $D \geq 1$,*

$$\boxed{|\mathfrak{X}_a| \leq \mathcal{B}_a(D)}. \tag{38}$$

For the literal erasure kernel the excluded site is already zero. If an enlarged bookkeeping kernel is used, removing that site changes only the $d = 1$ variation, by at most $4|\widehat{K}_a(b, a)|$.

Proof. The prefactor $\mu(a)\mu(b)$ in (8) has modulus one. Apply [theorem 5.1](#) and then [lemma 3.6](#). $\square$

7.2 The exact global trace certificate

Corollary 7.2 (Linear mixed-sector majorant). *For every choice of anchorwise cutoffs $D_a \geq 1$,*

$$|\text{tr}(ER)| \leq 2 \sum_{m < n} \sum_{a \in \mathcal{R}_{m,n}} \mathcal{B}_a(D_a), \tag{39}$$

$$|S_{2,1}| \leq 4 \sum_{m < n} \sum_{a \in \mathcal{R}_{m,n}} \mathcal{B}_a(D_a), \tag{40}$$

where $S_{2,1} = 2 \text{tr}(ER)$ is the full linear erasure coefficient in $\text{tr}(R + tE)^2$.

Proof. Use the exact anchor expansion (9), the triangle inequality, and [proposition 7.1](#). The second formula uses the definition of $S_{2,1}$. $\square$

Remark 7.3 (Normalization is already present). Every $K_{a,q}$ is built from a four-corner amplitude divided by $W_m W_n$. No further factor $\sqrt{W_m W_n}$ is introduced in (37). Doing so would double-normalize the rectangle scalar and would not turn it into an entry of E .

7.3 The three trace-level physical gates

[Corollary 7.2](#) separates the remaining work at the level it actually controls.

Gate A: low-diagonal variation. For every relevant anchor and every low d , one must control

$$\mathcal{V}_\square((K_a^\circ)^{[d]}).$$

The kernel contains the actual prime-ray incidence and every native mask. Replacing it by a smooth envelope proves a theorem for the envelope, not for the literal coefficient, unless the replacement error is separately bounded.

Gate B: diagonal tail relative to actual anchored mass. The ambient estimate

$$\mathcal{T}_D(K_a^\circ) \leq \|K_a^\circ\|_\infty U_a V_a / D$$

may be too large for a sparse carrier. It must be compared with an actual anchored reference scale, for example

$$\mathcal{A}_a = \sum_{u,v} |K_a^\circ(u, v)|,$$

or directly within the global physical trace ledger.

Gate C: growing-anchor summability. A saving for every fixed anchor does not imply a saving after summing the growing family

$$\bigcup_{m < n} \mathcal{R}_{m,n}.$$

One needs

$$\sum_{m < n} \sum_{\mathbf{a} \in \mathcal{R}_{m,n}} \mathcal{B}_{\mathbf{a}}(D_{\mathbf{a}}) = o(\mathcal{A}_{\text{tr}})$$

at the declared trace reference scale $\mathcal{A}_{\text{tr}}$. Fixed-anchor little-oh notation without uniform domination is insufficient.

7.4 Boundary, long-ray interior, and short-ray tail

The diagonal tail $\mathcal{T}_D$ should not be confused with the geometric short-ray tail. For each anchor, split the primitive support disjointly into

$$\Omega_{\text{bd}} \sqcup \Omega_{\text{int}} \sqcup \Omega_{\text{short}}.$$

Here Ω_{bd} contains small u or small v ; Ω_{int} contains cells on which both cofactor coordinates grow and the affine ray has many admissible time positions; and Ω_{short} contains directions whose ray length is $O(1)$. Let the corresponding zero-extended kernels be

$$K_{\text{bd}}, \quad K_{\text{int}}, \quad K_{\text{short}}.$$

Then the exact triangle decomposition gives

$$|\mathcal{S}(K_{\mathbf{a}}^{\circ})| \leq \|K_{\text{bd}}\|_{\ell^1} + \|K_{\text{short}}\|_{\ell^1} + \mathcal{B}_D(K_{\text{int}}). \quad (41)$$

Thus a sufficient anchored closure consists of

$$\|K_{\text{bd}}\|_{\ell^1} + \|K_{\text{short}}\|_{\ell^1} = o(\mathcal{A}_{\mathbf{a}})$$

together with a mass-normalized variation–tail certificate for the interior. Having only one time point on each short ray does not make the sum over all such directions small; its literal total mass must be estimated.

If only one of u, v is large in an intermediate sector, the anisotropic certificate [proposition 5.9](#) can be applied instead of discarding the sector. Its one-direction variation must still be verified for the complete hard mask.

7.5 Why the q -sum is not a fourth sign source

The common erasure content $q \mid |H|$ disappears from (5). It remains inside $K_{\mathbf{a},q}$ through ray geometry and weights. Therefore:

- cancellation among q -slices can occur only through their literal complex amplitudes;
- applying the triangle inequality before (6) may lose that cancellation; and
- the divisor bound $\tau(|H|) = |H|^{o(1)}$ controls only multiplicity, not Möbius parity.

7.6 The trace-to-operator bridge is open

A small value of $\text{tr}(ER)$ is not a norm bound for E , for $R + E$, or for the missing-native Gram. Global cancellation among anchors can coexist with large row-local defects. In particular, [proposition 7.1](#) gives no entrywise domination of E , so the star and Perron certificates of TPC–68 cannot be invoked directly.

An operator-level continuation needs at least one genuinely new bridge:

- a direct primitive reindexing of the two-corner entries $E_{m,n}$;
- row-local anchored sums that dominate the relevant stars;
- or uniform control of complete growing mixed moments, without reducing them to the single global trace.

This is an L2 interface, not a consequence of two-dimensional Abel summation.

7.7 Reassembly remains separate

Even an operator-level missing-native bound would control only one branch. The same-residual defect, total missing mass, represented lower frame, and the nuisance shorting term remain separate. In the full reassembly notation of TPC-68,

$$G_{\text{sh}} = G_{\text{dir}} - C_N.$$

No plane estimate implies $\|C_N\| = o(1)$, and no missing-native estimate supplies the represented lower bound. These arrows must be verified on the same carrier and with the same normalization.

8 Countermodels and kill criteria

The following examples prevent the regular-kernel theorem from being silently promoted to the literal carrier.

8.1 Bounded amplitude is insufficient

Define on $[1, U] \times [1, V]$

$$K_{\text{match}}(u, v) = \mu(uv).$$

Then $\|K_{\text{match}}\|_{\infty} \leq 1$, but

$$\mathcal{S}(K_{\text{match}}) = \sum_{u \leq U, v \leq V} \mu(uv)^2 = \mathfrak{S}_{\text{sf}} UV + o(UV). \quad (42)$$

Thus no estimate of the form $o(UV)$ can hold for all bounded kernels. The kernel has selected the exact arithmetic sign.

There is also a nonnegative version:

$$K_+(u, v) = \mathbf{1}_{\mu(uv)=1}.$$

By (36) and (33),

$$\mathcal{S}(K_+) = \frac{\mathfrak{S}_{\text{sf}}}{2} UV + o(UV). \quad (43)$$

Hence nonnegativity of the mask does not prevent coherent parity selection. These are abstract plane kernels; we do not assert that the physical prime-line mask equals either one.

8.2 A prime mask can freeze the plane sign

Let

$$K_{\mathbb{P}, N}(u, v) = \mathbf{1}_{\substack{u, v \leq N \\ u \neq v}} \text{ prime}.$$

Its support is primitive and squarefree. Since $\mu(p) = -1$ for every prime,

$$\mathcal{S}(K_{\mathbb{P}, N}) = \pi(N)(\pi(N) - 1) = \|K_{\mathbb{P}, N}\|_{\ell^1}.$$

Thus prime support by itself does not create Möbius cancellation; it can freeze the sign completely. This example is not the literal two-prime-line kernel, but it explains why an upper-bound sieve for its capacity cannot serve as a variation or discrepancy theorem.

8.3 A primitive line contains a Chowla-type boundary

Take

$$K_{\text{line}, N}(u, v) = \mathbf{1}_{v=u+1, u \leq N}.$$

Every supporting pair is coprime, but

$$\mathcal{S}(K_{\text{line}, N}) = \sum_{u \leq N} \mu(u)\mu(u+1).$$

An unweighted $o(N)$ theorem here is the open two-point Chowla correlation. The regular-kernel theorem correctly does not apply: a thin diagonal graph has mixed variation growing like its line length, rather than $O(\|K\|_\infty)$. Known logarithmically averaged results [4] do not turn this raw fixed line into an unweighted theorem.

8.4 Small ambient bound can be physically vacuous

Let K be supported on a set of size $A = o(UV)$, with $|K| \leq 1$. The unconditional conclusion

$$\mathcal{S}(K) = o(UV)$$

does not imply $o(A)$, and may be larger than the entire physical mass. The only relevant statement is the relative certificate

$$\mathcal{B}_D(K) = o(\mathcal{A}(K))$$

with $\mathcal{A}(K)$ chosen from the actual normalization. This is why [definition 5.4](#) is not cosmetic.

8.5 Pairwise decay can lose to row growth

For each dimension R , let a symmetric defect have zero diagonal and all off-diagonal entries R^{-1} . Every fixed entry tends to zero, but the top eigenvalue is

$$\frac{R-1}{R} \rightarrow 1.$$

Therefore a fixed-pair primitive-plane estimate, even if uniform in all other parameters, is not an operator saving unless its row sum or Perron radius also tends to zero.

8.6 A mixed trace does not control the erasure operator

Let

$$R = 0, \quad E = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then every ER rectangle scalar and $\text{tr}(ER)$ vanish, while

$$\|E\| = 1.$$

Thus even perfect cancellation in the anchored TPC-70 target cannot be renamed as an erasure-operator bound. A direct two-corner or complete moment bridge is indispensable.

8.7 Separate q -bounds can destroy cancellation

Take two divisor slices with

$$K_{q_1} = F, \quad K_{q_2} = -F.$$

The aggregate kernel is zero, while

$$\mathcal{V}_\square(K_{q_1}) + \mathcal{V}_\square(K_{q_2}) = 2\mathcal{V}_\square(F).$$

Thus the q -resolved triangle inequality can be strictly weaker than variation of the exact aggregate. Conversely, no cancellation between q -slices may be assumed unless their literal phases produce it.

8.8 Failure of one certificate is not failure of cancellation

Let K be any highly oscillatory kernel whose signed plane sum happens to vanish exactly. Its mixed variation can still be of order UV . Therefore a large value of $\mathcal{V}_\square(K^{[d]})$ kills the divisor-variation route only. It does not prove that the original physical coefficient is large.

8.9 Explicit stop rules

For the current route, the following outcomes are decisive.

K1. If the exact physical kernel has

$$\mathcal{B}_D(K_a^\circ) \neq o(\mathcal{A}_a)$$

for an anchor family carrying a nonnegligible part of the declared trace scale, then the present anchored variation route does not close.

K2. Even if the anchored sums make $\text{tr}(ER) = o(1)$ at its declared scale, the operator route stops unless a new direct-entry, row-local, or growing-moment bridge is proved. Global trace cancellation alone is not such a bridge.

K3. If the native defect becomes small but the mass comparison, represented lower frame, or nuisance shorting term fails, reassembly stops; the plane result is not upgraded to a prime-pair statement.

K4. If a required net exponent loss reaches or exceeds $1/400$, including equality, the current endpoint implication stops.

Each stop rule leaves a valid negative or boundary result: it identifies which exact certificate fails without claiming that every possible route fails.

9 Relation to existing Möbius-correlation results

The present theorem uses only one-variable Mertens cancellation after a full two-dimensional average and a diagonal divisor lift. It should not be confused with a theorem for a prescribed pair of affine forms.

For fixed nonproportional affine forms, logarithmically averaged two-point Chowla is known [4]. Averaged unweighted results are also available when an additional shift or coefficient is averaged [3]. Those theorems are powerful but do not directly control the literal kernel here, whose coefficients can grow with the physical scale and whose prime-line and native-key masks select a restricted subset of the cofactor plane.

TPC-33 records the detailed mismatch [5]. The unresolved physical requirements include:

- (i) growing affine coefficients and row-dependent primitive coordinates;
- (ii) a fixed dyadic physical shell rather than only logarithmic averaging;
- (iii) complete prime-line, endpoint, and native-key masks;
- (iv) actual complex weights and row normalization;
- (v) uniformity over all active row pairs; and
- (vi) an operator-scale saving compatible with the endpoint ledger.

Theorem 6.1 avoids the fixed-form Chowla barrier because it assumes regularity in two independent cofactor directions. If the kernel collapses onto a thin prescribed affine graph, its zero-extended mixed variation can be as large as its length, and the theorem does not manufacture a saving. This is the correct behavior: the family of prescribed affine directions contains open two-point Möbius correlations.

The classical parity obstruction for sieve methods remains relevant [1]. The divisor lift is not a parity-breaking sieve. It is an exact reindexing followed by prime-number-theorem cancellation under a deterministic regularity hypothesis. Whether the literal physical kernel satisfies that hypothesis is precisely the new arithmetic gate.

10 Claim ledger and next decisions

10.1 What is proved

L0: finite and analytic statements. The following results are unconditional:

- the primitive sign collapse $\mathbf{1}_{(u,v)=1}\mu(u)\mu(v) = \mu(uv)$;
- the exact diagonal divisor lift (13);
- the zero-extended two-dimensional Abel identity (16);
- fixed- d restricted Mertens cancellation from the prime number theorem;
- the exact master certificate (25) and ambient tail (26);
- the anisotropic one-direction certificates and the exact boundary/interior/short-ray decomposition;
- unconditional $o(UV)$ cancellation for kernels whose fixed-d mixed variation is $O_d(\|K\|_\infty)$; and
- the projective-BV and Banach-valued certificates.

L1: fixed-shift identification. On the complete squarefree physical carrier inherited from TPC-70, (8) identifies every anchored cross-key rectangle sum $\mathfrak{X}_a$ with the primitive-plane transform of the complete aggregate kernel K_a° . Their exact sum gives $\text{tr}(ER)$. The common erasure content q is sign-free, the sign is frozen on every affine time ray, and the same-residual direction is one excluded primitive-plane site, already zero in the literal erasure kernel. Combining this L1 dictionary with the L0 analytic theorem gives the anchored and global mixed-trace majorants (38)–(40).

10.2 What remains L2-open

No theorem in this paper proves:

- (1) bounded or saving mixed variation for the actual prime-line kernel;
- (2) a diagonal tail small relative to actual atomic mass;
- (3) uniform restricted-Mertens input for a growing divisor cutoff;
- (4) summability of anchored certificates at the physical trace scale;
- (5) a direct-entry, row-local, or growing-moment bridge from the anchored trace problem to an operator bound;
- (6) control of the same-residual part of the native defect;
- (7) a fixed-power missing-mass saving;
- (8) the represented activated-range lower frame;
- (9) a small nuisance shorting penalty; or
- (10) the boundary and tail estimates needed for the final physical implication.

10.3 Endpoint ledger

Any future composition must use one denominator-compatible ledger. A schematic safe form is

$$\Lambda_{\text{tot}} = \lambda_{\text{sel}} + \lambda_{\text{occ}} + \lambda_{\text{blk}} + \lambda_{\text{term}}^{\text{eff}} + \lambda_{\text{sing}} + \lambda_H + \lambda_{\text{aff}} + \lambda_{\text{reasm}} + \lambda_{\text{bdry}} + \lambda_{\text{tail}} < \frac{1}{400}.$$

The effective terminal loss already contains whichever inflation or capacity route is used; it must not be charged twice. Subpower divisor counts and a bounded number of exact relabelings cost zero exponent, but any polynomial variation, row-aggregation, or tail loss must be entered literally. Equality at $1/400$ is failure for the strict endpoint.

10.4 Next-step decision tree

The present paper leaves four testable branches.

- (I) **Direct physical variation.** Compute $\Delta_{12}(K_a^\circ)^{[d]}$ from the literal ray masks. If the jumps telescope or are confined to a small boundary, feed the resulting majorant into (39).
- (II) **Smooth plus discrepancy.** Decompose

$$K^\circ = K_{\text{reg}} + K_{\text{rough}}.$$

Apply the proved theorem to K_{reg} , and formulate a separate arithmetic discrepancy estimate for K_{rough} . Both parts must retain the actual normalization.

- (III) **Trace-to-operator bridge.** Construct either a direct two-corner primitive kernel for entries of E , a row-local anchor certificate, or a complete growing-moment estimate. A small global $\text{tr}(ER)$ is not enough.
- (IV) **Certificate failure.** If actual mixed variation is too large, record the exact offending masks and abandon this certificate. Possible alternatives include a direct affine-correlation estimate, a multi-modulus dispersion, or a growing-moment route. Failure of the BV certificate is not a no-cancellation theorem.

10.5 No endpoint claim

The paper does not specialize h_0 to 2, does not improve a fixed-shift parity estimate, and proves no prime-pair lower bound. In particular, it does not prove or imply the twin-prime conjecture. The genuine advance is narrower: the paper identifies an unconditionally cancellative regular comparison class and the explicit physical variation, tail, anchor-summability, and trace-to-operator gates needed to connect the literal kernel to that class.

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